Generative Image Denoising Using Conditional Diffusion Model

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***Abstract*—This project explores a Conditional Denoising Diffusion Probabilistic Model (DDPM) for image denoising, focusing on improving image quality by removing various types of synthetic noise. Traditional denoising methods such as CNNs and autoencoders struggle to preserve structural and textural details, often resulting in over-smoothed outputs. Our goal is to develop a conditional diffusion-based denoiser capable of reconstructing high-quality images with superior PSNR, SSIM, and perceptual quality metrics. Using the DIV2K dataset, we implemented a U-Net-based diffusion architecture and evaluated its performance against multiple types of degradation. Final experiments demonstrate promising results, achieving SSIM up to 0.779, PSNR of 24.47 dB, and LPIPS of 0.1730 at the best epoch. The findings suggest diffusion models provide a strong foundation for advanced generative image restoration tasks.**

***Index Terms*—Diffusion models, image denoising, DDPM, U-Net, PSNR, SSIM, LPIPS.**

I. Introduction

Image denoising is a long-standing problem in low-level computer vision that aims to recover clean images from noisy observations while preserving structural and textural details. Classical approaches such as Non-Local Means, BM3D, and total variation minimization rely on handcrafted priors and statistical assumptions, which often fail to generalize to complex real-world noise. Deep learning–based methods, including DnCNN and its successors, have significantly improved performance by learning powerful data-driven mappings between noisy and clean images. However, these deterministic models struggle to represent the full uncertainty inherent in the noise distribution.

Diffusion Probabilistic Models (DPMs) have recently emerged as a powerful class of generative models that learn to reverse a gradual noising process through a sequence of denoising steps. Unlike conventional deterministic networks, DPMs explicitly model the data distribution and progressively refine samples from pure noise to realistic images. This stochastic formulation allows them to capture complex data priors, leading to state-of-the-art results in generative synthesis and image restoration tasks such as super-resolution and inpainting. Despite these advances, two critical limitations restrict their use in practical denoising: the high computational cost resulting from hundreds of reverse diffusion steps, and an inherent frequency bias that causes over-smoothing and the loss of high-frequency details.

II. Literature Review

Probabilistic distribution based generative models progressively handle noise, and aim to closely match the probability distribution of the training data. DDMP model comprises two major processes i.e. the diffusion process also known as forward process and the reverse process [Ho et al., 2020]. The traditional method requires task-specific knowledge to model the degradation and perform restoration in the spatial or frequency domain, by combining classical signal algorithms with specific image degradation parameters.

Most deep learning models directly train neural networks on a set of paired LQ-HQ (Lower Quality-Higher Quality) images with a reconstruction objective, as typical in supervised learning. This approach tends to produce over-smooth results, particularly in textures. Although this issue can be alleviated by including adversarial or perceptual losses, the training typically becomes unstable, and the results often contain undesired artifacts when processing the input LQ images [Bhattacharya et al., 2023]. The forward process usually introduces Gaussian noise, $\epsilon \sim \mathcal{N}(0, I)$, to the initial input data $x_0 \sim q(x_0)$, which gradually diffuses the data into random noise. It also allows for the direct generation of data x_t at any time step t based on the initial data x_0, representing a linear combination of x_0 and the added noise ϵ [Choi et al., 2023].

$$q(x_t \mid x_{t-1}) = \mathcal{N}\Big(x_t; \sqrt{1-\beta_t}\, x_{t-1},\, \beta_t I\Big) \tag{1}$$

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^{T} q(x_t \mid x_{t-1}) \tag{2}$$

Here, x_t is obtained by adding Gaussian noise with variance β_t at timestamp t and scaling the previous sample x_{t-1} with $\sqrt{1-\beta_t}$ according to a variance schedule.

Accordingly, Reverse process aims to achieve the goal opposite to the forward process by denoising to generate a sample. We start with an initial random noise sample $x_t \sim \mathcal{N}(0, I)$ and progressively reduce the noise and gradually restore a sample. The reverse process learns to generate new data samples starting from the Gaussian noise.

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\big(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)\big) \tag{3}$$

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t) \tag{4}$$

III. Methodology

A. Data Preprocessing

We used a high-quality image dataset (DIV2K) with a variety of natural scenes. All images were resized or cropped to 256×256 pixels and processed as RGB inputs. Synthetic degradations added to clean images include:

- Image Resizing/Cropping: All images are resized or cropped to a fixed resolu- tion i.e 256 x 256 pixels,
- Synthetic Noise Generation: Multiple types of noise are added to clean images such as: Gaussian noise with (0.05, 0.1, 0.2) levels, JPEG compression with (90, 70, 50) quality factors and Motion Blur (1, 3, 5) levels.

Training/validation splits: 800 images for training and 100 images reserved for testing.

B. Model Architecture and Training

We implemented a conditional DDPM with a U-Net backbone that predicts the Gaussian noise at each timestep. Inputs are the noisy image x_t, conditioning y (if available), and timestep embedding t. The model predicts $\hat{\epsilon}(x_t, y, t)$ and is trained with the simplified MSE noise-prediction loss:

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{x_0,\epsilon,t}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2\right]. \tag{5}$$

Hyperparameters and training setup:

- Framework: PyTorch
- Optimizer: Adam, learning rate 2×10^{-4}
- Loss: MSE
- Epochs: 100
- Diffusion steps: 1000
- Hardware: Google Colab (GPU)

IV. Experiment

A. Evaluation Metrics

We used three standard image quality metrics: PSNR (Peak Signal-to-Noise Ratio) to measure pixel-level fidelity, where PSNR > 30 dB = excellent quality. Also used SSIM (Structural Similarity Index) to evaluate structural and contrast similarity (0–1 range). Similarly, LPIPS (Learned Perceptual Image Patch Similarity) to measure perceptual realism.

Epoch	Train Loss	PSNR (dB)	SSIM	LPIPS
12	0.0338	10.99	0.158	0.673
14	0.0330	17.86	0.606	0.4001
85	0.0307	24.47	0.779	0.173
100	0.0279	16.28	0.468	0.456

Fig. 1: Epoch table with training loss and other metrics

Metric	Inference Value	Best Training (Epoch 85)	Gap
PSNR	21.25 dB	24.47 dB	-3.22 dB
SSIM	0.611	0.779	-0.168
LPIPS	0.3293	0.1730	+0.1563

Fig. 2: Inference values

B. Results

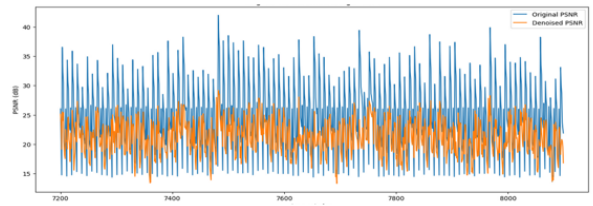

Fig. 3: Peak Signal-to-Noise Ratio (PSNR)

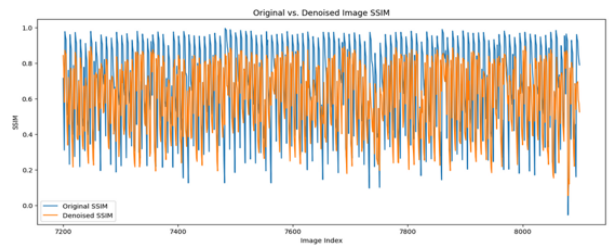

Fig. 4: Structural Similarity Index (SSIM)

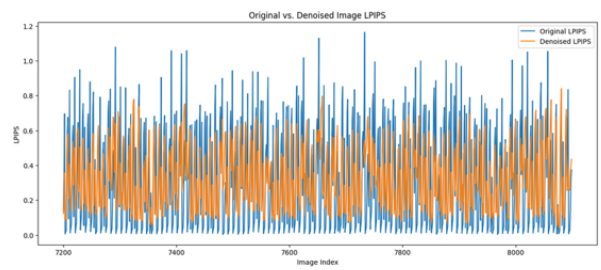

Fig. 5: Learned Perceptual Image Patch Similarity (LPIPS)

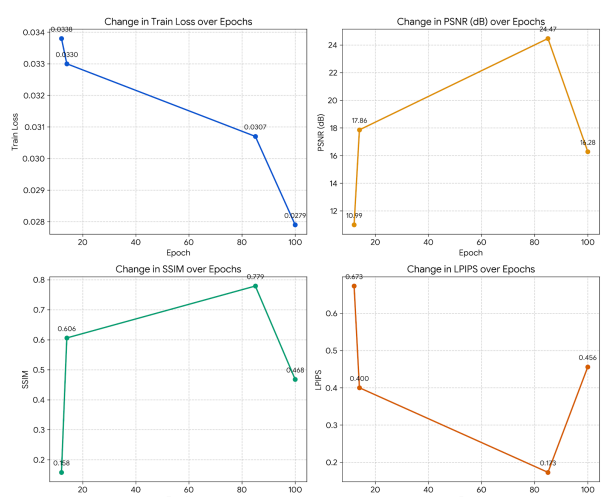

Fig. 6: Epoch performance metrics

In the figures, we can see that the performance of the model was evaluated using different metrics, including training loss, PSNR, SSIM, and LPIPS. The model achieved a training loss of 0.0307, representing the mean squared error between the

predicted noise and the actual noise at each timestep. A low training loss indicates that the model is effectively learning to denoise images. The PSNR value was 24.47 dB, which measures pixel-wise similarity between the reconstructed and ground truth images — higher PSNR indicates better reconstruction. The SSIM value of 0.779 reflects the model's ability to preserve structural information and overall image layout.

V. Discussion

The diffusion-based method achieves promising denoising performance and preserves structural details better than many deterministic baselines, though PSNR remains below the "excellent" threshold (30 dB) for the best epoch. The stochastic formulation enables a richer prior, but runtime cost from many reverse steps and occasional smoothing of high-frequency detail remain concerns. Future work includes experimenting with cosine variance schedules, fewer-step samplers, and perceptual/adversarial loss terms.

VI. Conclusion

We presented a conditional diffusion model for image denoising using a U-Net backbone trained on synthetically degraded data. The model attained PSNR of 24.47 dB, SSIM of 0.779, and LPIPS of 0.173 at the best epoch. Future steps include evaluation on real noisy datasets, improved samplers for inference speed, and additional loss terms to enhance perceptual realism.

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TABLE I: Team Contributions

Name	**Section & Level (Graduate)**	**Contribution**
Santosh Luitel	CSC 557 U20	• Worked on model training, evaluation, and tuning of diffusion parameters. • Analyzed experimental results and prepared performance comparisons (PSNR, SSIM, LPIPS).
Gaurav Regmi	CSC 557 U20	• Handled data preprocessing and augmentation. • Assisted in model implementation using U-Net architecture.
Aavash Silwal	CSC 557 U20	• Worked on preparing slides and designing the overall presentation layout. • Helped design the methodology section and worked on coding the diffusion and reverse process and training the model.
Sambhav Siwakoti	CSC 557 U20	• Focused and researched on diffusion models and their applications. • Helped design the methodology section and documented the diffusion and reverse process.